Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/22/2007
PI Name:	Morris, Jeffrey F. Ph.D.		
Project Title:	Droplet formation processes from particle-laden liquids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Professor F. Joseph Schork has acted as PI at Georgia Tech, while Dr. Jeff Morris (who was at Halliburton) remained technically responsible for the project. The grant is being transferred to CCNY where Dr Morris is joining the faculty January 1, 2005.		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	04/01/2002	**End Date:**	01/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date renegotiated to 1/31/2007, per PI (3/2007)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC05GA13G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The study addresses basic drop formation processes for the dripping (or pendant drop) mode and laminar jet instability for solids-laden liquids, processes previously intensively studied for pure liquids. This work is building from that knowledge base by experimentally examining drop formation processes of noncolloidal suspensions at particle volume fractions up to, to date, 40% and eventually 50%. The tools used include high-speed video imaging and image analysis, with analytical tools from suspension mechanics and stability of fluid flow applied to the interpretation of the behavior. The primary issues are to determine the role of particle fraction and particle size, relative to the orifice diameter, in the drop formation process. The study is designed to isolate bulk rheological effects from finite particle size effects, in support of the goals of establishing the similarities and differences of the drop formation process and the associated fluid mechanical instabilities in a suspension of solid particles in relation to the equivalent processes in pure liquids. Numerical analysis using singularity methods to describe the particle effects on-axis in the jet are proposed for study. The basic physical processes in drop formation from mixtures are unexplored, while the phenomena are found in a number of applications. Thus the significance includes fundamental development of suspension mechanics in a free-surface flow with known instabilities, and also will provide guidance to the development of technologies for solid application through drop location processes. The most prevalent of these are inkjet processes in the electronics and textile industries, where the goal is to apply solids in fine lines for circuitry and imaging, respectively. Particle-laden "inks" raise issues in both cases because the very rapid pinch of small drops leads to a situation where the influences of rheology and isolated particle effects are largely unknown. Increased knowledge thus promises to open the way to new scientific approaches and at the same time the elucidation of basic processes should rapidly aid in the design of inkjet applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The benefits of the research toward technologies on earth can be expected in coating and inkjet type of applications. The basic science of mixture flows with free surfaces is generally poor, and is especially so when there is a rupture event such as drop formation involved: the presence of the particles renders the process significantly irregular and thus requires not only a reexamination using suspension mechanical concepts, but also statistical concepts. The advances made here will support understanding of inkjet printing with, for example, metal-laden liquids or ceramic particles, technologies under examination (largely by empirical approaches) for a range of applications.
Task Progress:	The year's activities include two experimental thrusts and a novel numerical thrust. The experiments have focused on moving toward the smaller scale systems encountered in application, while the numerical approach is based on coupling of two well-established techniques, and is valuable for providing great detail on a model representation of the drop-formation process. We first note a major effort to develop a technique capable of capturing drop formation of application-scale drops in drop-on-demand inkjet systems was made. The goal is to obtain the spatial and temporal resolution to allow us to assess the role suspended colloidal particulates have on the process. Using a stroboscopic approach which is a variant of classical methods but under much more rapid time, O(20 ns), and small space , O(100) micron per side, scales, a successful synchronization of a pulsed laser with a high-resolution camera and triggering of the drop formation allowed us to reconstruct the drop formation events with micro-second accuracy and submicron spatial resolution. This work is now in press for pure liquid systems, and extension to particle-laden systems is underway. A second experimental effort is to examine particle-laden jet breakup in continuous mode for jets of O(100) micron initial diameter. The role of particle loading, both in terms of particle size and volume fraction are explored. The goal is to compare the results of this study with the detailed results for larger (millimeter scale) drops in order to provide insight to the role of capillarity associated with particle-induced deformation. In a related study, a controlled introduction of particles to the cenetrline a jet by a syringe-within-a-syringe technique is under investigation. This allows us to precisely control the location of the solids to be on-axis with the jet, and thus provides a system much more readily compared with the numerical techniques described below. A high-school student is working with the post-doctoral researcher and the PI on the experiments and will continue the project over the next year and describe the work as part of a national science competition. The numerical analysis of a stretching jet, both with and without an included particle, under viscous flow conditions has been studied. Having successfully reproduced literature results for a particle-free jet thinning toward pinch for unstretched and stretched conditions, we moved to an examination in which a particle placed on-axis within a column of fluid is represented as a force dipole. The approach successfully mimics behavior seen in dripping mode drop formation in earlier parts of this study. We have allowed the particle to be mobile in an unstretched but thinning column of fluid, enforcing force-free conditions on the body. Comparison of the particle motion to cases seen in experiment indicates the approach is valid for early stage behavior, and the case of a stretching jet containing multiple particles on-axis was also examined. Publication of this work is in preparation. The major accomplishments include also publication of results of work previously outlined, including work showing the manner in which particle loading alters the breakup structure based on a two-stage analysis of the process. Work is in progress to understand the near-orifice structure of a slow jet by a combination of a simplified model of the particles and experiments. The experimentally-determined behavior shows that the particles alter the curvature of the jet near the orifice, and are believed to be a uniquely two-phase behavior.
Bibliography Type:	Description: (Last Updated: 06/25/2007)
Articles in Peer-reviewed Journals	Furbank RJ, Morris JF. "Pendant drop thread dynamics and pinch-off in particle-laden liquids." International Journal of Multiphase Flow in press, April 2006. , Apr-2006
Books/Book Chapters	Park H, Ok H, Carr WW, Furbank RJ, Morris JF. "Drop formation and impaction." in "Digital printing of textiles." Ed. H. Ujiie. Cambridge : Woodhead Publishing Limited, 2006. ISBN 1 85573 951 8., Apr-2006